

Wencheng XIA

Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore

E-mail: wencheng.xia@ntu.edu.sg

Education Background

B.S. in Bioengineering

School of Life Science, Beijing Institute of Technology, China

09/2014-07/2018

Research Assistant

Interdisciplinary Research Center on Biology and Chemistry, SIOC, Chinese Academy of Sciences, China

03/2019-08/2020

Ph.D. in Cell Biology

Interdisciplinary Research Center on Biology and Chemistry, SIOC, Chinese Academy of Sciences, China

09/2020-07/2025

Research Focus: Cryo-EM studies of protein fibrils associated with neurodegeneration

Supervisor: Prof. Cong LIU

Postdoctoral Research Fellow

Lee Kong Chian School of Medicine, Nanyang Technological University

04/2026-Present

Research Focus: Cryo-EM and cryo-ET studies of membrane proteins and vesicles associated with neurodegeneration

Supervisor: Asst Prof. Chuchu WANG

Major Research

Research Interest:

Neurodegenerative diseases, Structure-Function-Application study, Cryo-EM, Cryo-ET, Molecular Mechanism and Interaction Analysis, Amyloid Fibril, Post-translational Modifications (PTMs), Membrane Protein and Vesicles.

1. Alzheimers disease -related glycosylation of A β fibril:

- Glycosylation modification drives A β ₄₂ to adopt a fibril conformation that is distinct from the wild types, with a clear additional electron density corresponding to the glycan at Y10;
- Glycosylated A β ₄₂ fibrils form a less stable protofilament interface compared to the wild types;
- Glycosylated A β ₉₋₂₁ induces A β ₄₂ to adopt three entirely new fibril conformations;
- The glycosylated A β ₉₋₂₁ not only contributes to the formation of the fibril core, but also interacts with adjacent peptide chains via a zipper-like mechanism to collaboratively stabilize the fibril structures.

2. Parkinsons disease -related glycosylation and phosphorylation of α -synuclein:

- Both glycosylated and phosphorylated α -syn fibrils presented a reduced capability for fibrillation;
- Cryo-EM further confirm that both PTMs lead to entirely novel fibril conformations compared to wild-type;
- Both glycosylation and phosphorylation reduce the aggregation propensity of α -syn and significantly diminish the propagation and neurotoxicity of α -syn fibrils.

3. Glycosylation regulated peptide functional amyloid fibrillation:

- The glycosylation induces peptides to self-assemble into highly ordered and uniformly shaped fibrils and glycan play a pivotal role in aggregation process;
- Glycopeptide fibrils adopt a typical cross- β structure with a “triple-helix” conformation and intermolecular interactions collectively stabilize the overall fibril conformation;

- The glycopeptide fibril promotes cell proliferation, cell migration, and osteogenic differentiation.

Publication

Co-first author (#)

- 1) **Xia W.**[#], Xu Z.[#], Dong H., Zhang S., He C., Li D., Sun B., Dai B., Dong S., Liu C. Design and Structural Elucidation of Glycopeptide Fibrils: Emulating Glycosaminoglycan Functions for Biomedical Applications. *J Am Chem Soc.* 2025.
- 2) Zeng S.[#], Zhang S.[#], Zhang S.[#], Fan Y.[#], **Xia W.**[#], Chen F.[#], Huang C., Lv S., Lu J., Sun Y., Liu K., Li Y., Zhang Y., Wang J., Liu C., Li D. TPPP/p25 amyloid seeding activity as a specific biomarker for multiple system atrophy. *Cell.* 2026
- 3) Wang M.[#], **Xia W.**[#], Zhao D., Zhai Z., Chen R., Bai X., Zhang Z., Fan H., Zhang JP., Liu C., Jiao F. Amyloid Fibrillation of a Ninjurin-1-Derived α -Helical Peptide: Structural Insights into Conformational Transition. *ACS Nano.* 2025
- 4) Wei Q.[#], Liu D.[#], **Xia W.**[#], Wang F.[#], Liu Y., Liu C., Dong S. An O-glycopeptide Participates in the Formation of Novel A β 42 Fibril Structures and Attenuates A β 42 Neurotoxicity. *Nat Commun.* 2025.
- 5) Han Y.[#], Li J.[#], **Xia W.**[#], Li Q., Sun Z., Zeng W., Hu Y., Luk K., Liu C., Xiang S., He Z. Fibril fuzzy coat is important for α -synuclein pathological transmission activity. *Neuron.* 2025.
- 6) Hu J.[#], **Xia W.**[#], Zeng S.[#], Lim Y.[#], Tao Y., Sun Y., Zhao L., Wang H., Le W., Li D., Zhang S., Liu C., Li Y. Phosphorylation and O-GlcNAcylation at the same α -synuclein site generate distinct fibril structures. *Nat Commun.* 2024.
- 7) Wang F.[#], **Xia W.**[#], Zhang M., Wu R., Song X., Hao Y., Feng Y., Zhang L., Li D., Kang W., Liu C., Liu L. Engineering of antimicrobial peptide fibrils with feedback degradation of bacterial-secreted enzymes. *Chem Sci.* 2023.
- 8) Tao Y.[#], **Xia W.**[#], Zhao Q.[#], Xiang H., Han C., Zhang S., Gu W., Tang W., Li Y., Tan L., Li D., Liu C. Structural mechanism for specific binding of chemical compounds to amyloid fibrils. *Nat Chem Biol.* 2023.
- 9) Fan Y.[#], Zhao Q.[#], **Xia W.**[#], Tao Y., Yu W., Chen M., Liu Y., Zhao J., Shen Y., Sun Y., Si C., Zhang S., Zhang Y., Li W., Liu C., Wang J., Li D. Generic amyloid fibrillation of TMEM106B in patient with Parkinsons disease dementia and normal elders. *Cell Res.* 2022.
- 10) Liu D.[#], Wei Q.[#], **Xia W.**[#], He C., Zhang Q., Huang L., Wang X., Sun Y., Ma Y., Zhang X., Wang Y., Shi X., Liu C., Dong S. O-Glycosylation Induces Amyloid- β To Form New Fibril Polymorphs Vulnerable for Degradation. *J Am Chem Soc.* 2021.

Co-author

- 1) Single-Molecule Insight Into α -Synuclein Fibril Structure and Mechanics Modulated by Chemical Compounds. *Adv Sci (Weinh).* 2025.
- 2) A stable and biocompatible shortwave infrared nanoribbon for dual-channel in vivo imaging. *Nat Commun.* 2025.
- 3) Amyloid fibril structures and ferroptosis activation induced by ALS-causing SOD1 mutations. *Sci Adv.* 2024.
- 4) Binding adaptability of chemical ligands to polymorphic α -synuclein amyloid fibrils. *Proc Natl Acad Sci U S A.* 2024.
- 5) Ultrasensitive detection of aggregated α -synuclein using quiescent seed amplification assay for the diagnosis of Parkinsons disease. *Transl Neurodegener.* 2024.
- 6) A Tau PET tracer PBB3 binds to TMEM106B amyloid fibril in brain. *Cell Discov.* 2024.
- 7) Creating an Amyloid Kaleidoscope Using Short Iodinated Peptides. *Angew Chem Int Ed Engl.* 2023.
- 8) Development of an α -synuclein positron emission tomography tracer for imaging synucleinopathies. *Cell.* 2023.
- 9) Conformational Dynamics of an α -Synuclein Fibril upon Receptor Binding Revealed by Insensitive Nuclei Enhanced by Polarization Transfer-Based Solid-State Nuclear Magnetic Resonance and Cryo-Electron Microscopy. *J Am Chem Soc.* 2023.
- 10) Activating NO-sGC crosstalk in the mouse vascular niche promotes vascular integrity and mitigates acute lung injury. *J Exp Med.* 2023.
- 11) Conformational change of α -synuclein fibrils in cerebrospinal fluid from different clinical phases of Parkinsons disease. *Structure.* 2023.
- 12) Subtle change of fibrillation condition leads to substantial alteration of recombinant Tau fibril structure. *iScience.* 2022.
- 13) SARS-CoV-2 impairs the disassembly of stress granules and promotes ALS-associated amyloid aggregation. *Protein Cell.* 2022.
- 14) Heparin induces α -synuclein to form new fibril polymorphs with attenuated neuropathology. *Nat Commun.* 2022.
- 15) Liquid-liquid phase separation of RBGD2/4 is required for heat stress resistance in Arabidopsis. *Dev Cell.* 2022.
- 16) Molecular structure of an amyloid fibril formed by FUS low-complexity domain. *iScience.* 2021.
- 17) The hereditary mutation G51D unlocks a distinct fibril strain transmissible to wild-type α -synuclein. *Nat Commun.* 2021.
- 18) Wild-type α -synuclein inherits the structure and exacerbated neuropathology of E46K mutant fibril strain by cross-seeding. *Proc Natl Acad Sci U S A.* 2021.
- 19) The nuclear localization sequence mediates hnRNPA1 amyloid fibril formation revealed by cryoEM structure. *Nat Commun.* 2020.
- 20) Hsp40 proteins phase separate to chaperone the assembly and maintenance of membraneless organelles. *Proc Natl Acad Sci U S A.* 2020.